Sharp fractional-power intertwining inequalities

Matthew J. Colbrook
Department of Applied Mathematics and Theoretical Physics
University of Cambridge, United Kingdom
mjc249@maths.cam.ac.uk

3 November 2026

Abstract

We prove the fractional-power intertwining inequality with sharp constant one for positive matrices and all exponents between zero and one. The argument yields spectral-floor refinements and a logarithmic commutator bound, and extends in operator norm to bounded operators and continuous nonnegative operator-monotone functions.

Repository target. This manuscript concerns AIM problem 897 at revision `5d7e26e8444e`. The precise scope is stated below.

1 Main result and proof

The original fractional-power question is recorded in [5, Problem X.5.3].

Answer. The constant-one inequality holds for every exponent $0 < r < 1$. The main step is a sharp estimate for shifted square roots. A positive integral representation proves the result above exponent one half, and repeated square roots cover the remaining exponents.

Theorem 1.1. *For every integer $n \geq 1$, positive semidefinite matrices $A, B \in M_n(\mathbb{C})$, matrix $X \in M_n(\mathbb{C})$, and $0 < r < 1$,*

$$\boxed{\|A^r X - X B^r\| \leq \|X\|^{1-r} \|AX - XB\|^r.}$$

The norms are operator norms, and the constant one is optimal.

1.1 An estimate at an extremal commutator vector

Lemma 1.2. *Let $M \geq m_0 I$ with $m_0 \geq 0$, and let $K^* = -K$ be finite complex matrices. Put $k = \|K\|$, $H = [M, K] = MK - KM$, and $h = \|H\| > 0$. After replacing K by $-K$ if necessary, choose a unit vector w with $Hw = hw$. Then*

$$w^* M w \geq m_0 + \frac{h}{2k}. \tag{897.1}$$

Proof. The matrix H is self-adjoint and its spectrum lies in $[-h, h]$. A choice of sign ensures that h is an eigenvalue. With $P = ww^*$,

$$H \geq h(2P - I). \tag{897.2}$$

Indeed, both sides agree on the span of w, while on its orthogonal complement the assertion is $H \geq -hI$. Also $k > 0$, since $K = 0$ would imply $H = 0$.

Set $U_t = e^{tK}$ for real t. These matrices are unitary. The matrix $L = -iK$ is self-adjoint with spectrum in $[-k, k]$, and $U_t = e^{itL}$. For $|t| \leq \pi/(4k)$, spectral calculus gives

$$\operatorname{Re}(w^* U_t w) = w^* \cos(tL) w \geq \cos(k|t|) > 0.$$

It follows from (897.2) that

$$\begin{aligned} g(t) := w^* U_t^* H U_t w &\geq h\big(2|w^* U_t w|^2 - 1\big) \\ &\geq h \cos(2kt) \qquad \left(|t| \leq \frac{\pi}{4k}\right). \end{aligned} \tag{897.3}$$

The function $f(t) = w^* U_t^* M U_t w$ satisfies $f(t) \geq m_0$ and

$$f'(t) = w^* U_t^* (MK - KM) U_t w = g(t).$$

Integrating over the preceding quarter-period yields

$$\begin{aligned} w^* M w = f(0) &= f\left(-\frac{\pi}{4k}\right) + \int_{-\pi/(4k)}^{0} g(t)\, dt \\ &\geq m_0 + h \int_{-\pi/(4k)}^{0} \cos(2kt)\, dt = m_0 + \frac{h}{2k}. \end{aligned}$$

□

Proposition 1.3. *For $M \geq m_0 I \geq 0$ and skew-adjoint K with $k = \|K\| > 0$, put $d = \|[M^2, K]\|$. Then*

$$\|[M, K]\| \leq k\left(\sqrt{m_0^2 + d/k} - m_0\right). \tag{897.4}$$

Proof. If $h = \|[M, K]\| = 0$, the assertion is immediate. Otherwise use the sign choice and vector from Lemma 1.2. With $H = [M, K]$, the identity $[M^2, K] = MH + HM$ gives

$$d \geq \left|w^* [M^2, K] w\right| = 2h\, w^* M w \geq 2m_0 h + \frac{h^2}{k}.$$

Solving this quadratic inequality for $h \geq 0$ proves (897.4). The sign change of K leaves all norms unchanged. □

1.2 The shifted square-root inequality

Let $A, B \geq 0$, $X \neq 0$, and write

$$k = \|X\| > 0, \qquad d = \|AX - XB\|, \qquad \delta = d/k.$$

For every $s \geq 0$ we claim the sharper estimate

$$\|\sqrt{A + sI}\, X - X \sqrt{B + sI}\| \leq k(\sqrt{s + \delta} - \sqrt{s}). \tag{897.5}$$

To prove it, use the block matrices

$$M = \begin{pmatrix} \sqrt{A + sI} & 0 \\ 0 & \sqrt{B + sI} \end{pmatrix}, \qquad K = \begin{pmatrix} 0 & X \\ -X^* & 0 \end{pmatrix}.$$

Here $M \geq \sqrt{s}\, I$, $K^* = -K$, and $\|K\| = \|X\| = k$. Writing $Y = \sqrt{A + sI}\, X - X\sqrt{B + sI}$ and $Z = AX - XB$ gives

$$[M, K] = \begin{pmatrix} 0 & Y \\ Y^* & 0 \end{pmatrix}, \qquad [M^2, K] = \begin{pmatrix} 0 & Z \\ Z^* & 0 \end{pmatrix}.$$

Their norms are $\|Y\|$ and $\|Z\| = d$, respectively; the terms sX cancel in the second commutator. Proposition 1.3 with $m_0 = \sqrt{s}$ proves (897.5). At $s = 0$ it gives the desired square-root inequality

$$\|A^{1/2}X - XB^{1/2}\| \leq \|X\|^{1/2}\|AX - XB\|^{1/2}. \tag{897.6}$$

In particular, if $d = 0$, every shifted square-root commutator in (897.5) is zero.

1.3 A positive integral for exponents above one half

Fix $1/2 < r < 1$ and define

$$C_r^{-1} = \int_0^\infty (\sqrt{1+s} - \sqrt{s}) s^{r-3/2}\, ds. \tag{897.7}$$

This is a finite positive number. Near zero the integrand is bounded by $s^{r-3/2}$, whose exponent is greater than -1. At infinity it is bounded by $\frac{1}{2}s^{r-2}$, whose exponent is less than -1. Substituting $s = tu$ proves, for $t > 0$,

$$t^r = C_r \int_0^\infty (\sqrt{t+s} - \sqrt{s}) s^{r-3/2}\, ds. \tag{897.8}$$

The same identity holds at $t = 0$, where the integrand vanishes. If an explicit special-function expression is desired, the identity

$$\sqrt{1+s} - \sqrt{s} = \frac{1}{2}\int_0^1 (s+u)^{-1/2}\, du$$

and Tonelli's theorem give

$$C_r = \frac{2r}{\mathrm{B}(r - 1/2, 1 - r)} = \frac{2r\sqrt{\pi}}{\Gamma(r - 1/2)\Gamma(1 - r)}.$$

Only positivity and the normalization in (897.7) are needed below.

Formula (897.8) applies to positive matrices by spectral calculus, with convergence in operator norm. To verify the convergence explicitly, for $A \geq 0$ and $s > 0$,

$$\|\sqrt{A + sI} - \sqrt{s}\, I\| = \sqrt{\|A\| + s} - \sqrt{s} \leq \min\left\{\sqrt{\|A\|}, \frac{\|A\|}{2\sqrt{s}}\right\}.$$

The same integrable endpoint bounds used above therefore apply. Consequently the scalar terms $\sqrt{s}\, X$ cancel to give

$$A^r X - XB^r = C_r \int_0^\infty (\sqrt{A + sI}\, X - X\sqrt{B + sI}) s^{r-3/2}\, ds.$$

Using (897.5), positivity of the weight, and then (897.8) with $t = \delta$ yields

$$\begin{aligned} \|A^r X - XB^r\| &\leq kC_r \int_0^\infty (\sqrt{s+\delta} - \sqrt{s}) s^{r-3/2}\, ds \\ &= k\delta^r = \|X\|^{1-r}\|AX - XB\|^r. \end{aligned} \tag{897.9}$$

This also covers $d = 0$, because both the integrand bound and δ^r are then zero.

1.4 All remaining exponents and sharpness

An inequality with constant one at exponents α and β implies it at their product. Indeed,

$$\begin{aligned}\|A^{\alpha\beta}X - XB^{\alpha\beta}\| &\le \|X\|^{1-\beta}\|A^{\alpha}X - XB^{\alpha}\|^{\beta} \\ &\le \|X\|^{1-\alpha\beta}\|AX - XB\|^{\alpha\beta}.\end{aligned}$$

For any $0 < r \le 1/2$, choose an integer $j \ge 1$ such that $\alpha = 2^j r \in (1/2, 1]$. The exponent α is covered by (897.9), or is 1, for which the inequality is an equality. Compose it with j applications of (897.6) to obtain the inequality at $r = \alpha 2^{-j}$.

If $X = 0$, both sides of Theorem 1.1 are zero, so the case omitted when dividing by k is immediate. All $d = 0$ cases were included above. Finally, the one-dimensional choice $A = 1$, $B = 0$, $X = 1$ gives equality for every r, proving that the constant cannot be decreased.

Relation to the cited problems. Loring and Vides isolate the square-root constant-one target as Conjecture 1.6 in their versioned preprint [1]. Herrera's later work [2] retains that question while proving a factor below 1.00891 for square roots and below 1.01975 for the broader family. The shifted estimate (897.5) is the step that makes the positive integral argument exact. The resulting Theorem 1.1 has all the quantifiers of the catalogue statement [3].

Extension to bounded operators and unital C^*-algebras. The same inequalities hold in every unital C^*-algebra, with its C^*-norm. To remove the finite-dimensional eigenvector assumption, choose the sign of H so that $h = \|H\| \in \sigma(H)$ and a state φ with $\varphi(H) = h$. In its GNS representation (π, ξ), $\pi(H)\xi = h\xi$: indeed, $0 \le hI - H \le 2hI$ implies $\varphi((hI - H)^2) = 0$. The orbit proof applies to $\pi(M), \pi(K), \xi$ using the bounds $\|\pi(H)\| \le h$ and $\|\pi(K)\| \le k$, and gives $\varphi(M) \ge m_0 + h/(2k)$. Thus $\|[M^2, K]\| \ge |\varphi(MH + HM)| = 2h\varphi(M)$, exactly as before. The block matrices belong to $M_2(\mathcal{A})$, and all functional-calculus integrals converge in norm, so the rest of the proof is unchanged.

1.5 A sharp bound with a positive spectral lower bound

Corollary 1.4. *Suppose $A, B \ge aI$ with $a \ge 0$, and let $X \ne 0$. Write $k = \|X\|$ and $d = \|AX - XB\|$. Then, for every $0 < r < 1$,*

$$\|A^r X - XB^r\| \le k((a + d/k)^r - a^r). \tag{897.10}$$

If $a > 0$, one also has

$$\|(\log A)X - X \log B\| \le k \log\left(1 + \frac{d}{ak}\right). \tag{897.11}$$

Both bounds are sharp.

Proof. Put $\delta = d/k$. For the shifted square-root matrices used above, the improved lower bound is $M \ge \sqrt{a + s}\, I$. Thus Proposition 1.3 gives

$$\|\sqrt{A + sI}\, X - X\sqrt{B + sI}\| \le k(\sqrt{a + s + \delta} - \sqrt{a + s}).$$

For $1/2 < r < 1$, integrating this inequality with the same positive weight as in (897.8) gives exactly $k((a + \delta)^r - a^r)$. The case $r = 1/2$ follows by setting $s = 0$.

The refined estimate is also preserved under composition. If it holds at α, then $A^\alpha, B^\alpha \ge a^\alpha I$ and

$$\delta_\alpha := k^{-1}\|A^\alpha X - XB^\alpha\| \le (a + \delta)^\alpha - a^\alpha.$$

Applying the refined estimate at β and using scalar monotonicity therefore gives

$$k^{-1}\|A^{\alpha\beta}X - XB^{\alpha\beta}\| \le (a^\alpha + \delta_\alpha)^\beta - a^{\alpha\beta} \le (a+\delta)^{\alpha\beta} - a^{\alpha\beta}.$$

The same square-root iteration used in the theorem proves (897.10) for all remaining r.

If $a > 0$, divide (897.10) by r and let $r \downarrow 0$. Finite-dimensional spectral calculus gives convergence in operator norm of the left-hand side to the logarithmic commutator, and the scalar right-hand side tends to $k\log((a+\delta)/a)$. This proves (897.11). The one-dimensional choices $A = a + \delta$, $B = a$, and $X = 1$ give equality, with $a > 0$ in the logarithmic case. □

1.6 A real-matrix reduction used during the investigation

There is also a reduction of potential counterexamples to real matrices of the same order, independent of the proof above. Diagonalize A, B and set $D_{ij} = a_i - b_j$ and $F_{ij} = a_i^r - b_j^r$. The two commutators become $D \circ X$ and $F \circ X$, where $\circ$ denotes entrywise multiplication. If $F \circ X = 0$, there is no counterexample. Otherwise choose unit singular vectors u, v with $u^*(F \circ X)v = \|F \circ X\|$. Diagonal unitary matrices U, V can be chosen so that $u = U|u|$ and $v = V|v|$. For $Z = U^*XV$, all three relevant norms are unchanged, and

$$|u|^{\mathsf{T}}(F \circ Z)|v| = \|F \circ X\|.$$

Set $R = \operatorname{Re} Z = (Z + \overline{Z})/2$. Conjugation preserves the operator norm, and the multipliers D, F are real. Hence

$$\|R\| \le \|X\|, \qquad \|D \circ R\| \le \|D \circ X\|, \qquad \|F \circ R\| \le \|F \circ X\|.$$

The last inequality is an equality: taking real parts preserves the displayed singular-vector witness. Thus R preserves the numerator and cannot increase either factor on the right. Any complex counterexample would therefore yield a real one of the same order.

1.7 Extension to nonnegative operator-monotone functions

The shifted-square-root estimate also yields a broader consequence. Here the norm remains the operator norm throughout.

Theorem 1.5. *Let $f : [0,\infty) \to [0,\infty)$ be continuous and operator monotone. For positive elements A, B of a unital C^*-algebra and any element X, with $k = \|X\| > 0$ and $\delta = \|AX - XB\|/k$, one has*

$$\|f(A)X - Xf(B)\| \le k(f(\delta) - f(0)). \tag{897.12}$$

The case $X = 0$ has both sides zero. In particular, the weaker right-hand side $kf(\delta)$ also holds. The scalar choices $A = \delta$, $B = 0$, and $X = 1$ attain the displayed bound.

Proof. Call a continuous increasing function $g : [0,\infty) \to [0,\infty)$ with $g(0) = 0$ admissible if $\|g(A)X - Xg(B)\| \le kg(\delta)$ for every such A, B, X. Two admissible functions have admissible composition: applying the outer estimate to the inner function and using scalar monotonicity proves this directly. The class is also closed under nonnegative linear combinations and limits that are uniform on compact intervals, by continuous functional calculus and continuity of the norm.

For $h > 0$, the function

$$g_h(t) = \frac{\sqrt{1+4ht} - 1}{2h}$$

is admissible by (897.5), applied with shift 1 to $4hA, 4hB$. Fix $\tau > 0$ and set $h = \tau/n$. If $x_0 = t > 0$ and $x_j = g_h(x_{j-1})$, then

$$x_{j-1} = x_j + hx_j^2, \qquad 0 < x_j \le t, \qquad \frac{1}{x_j} - \frac{1}{x_{j-1}} = \frac{h}{1 + hx_j}.$$

Summing the bounds $h/(1 + ht) \le h/(1 + hx_j) \le h$ gives

$$\frac{t}{1 + \tau t} \le g_{\tau/n}^{\circ n}(t) \le \frac{t(1 + \tau t/n)}{1 + \tau t + \tau t/n}.$$

The same bounds hold at $t = 0$. The difference between the two endpoints is

$$\frac{\tau^2 t^3}{n(1 + \tau t)(1 + \tau t + \tau t/n)} \le \frac{\tau^2 R^3}{n} \qquad (0 \le t \le R).$$

Thus $t \mapsto t/(1+\tau t)$ is an admissible compact-uniform limit of admissible iterates. Taking $\tau = 1/s$ and multiplying by $1/s$ shows that

$$t \longmapsto \frac{t}{t + s} \quad \text{is admissible for every } s > 0.$$

The positive integral representation [4, Proposition 1.1], absorbing its factor $1 + s$ into the measure on $(0, \infty)$, gives

$$f(t) - f(0) = bt + \int_{(0,\infty)} \frac{t}{t + s}\, d\mu(s), \qquad b \ge 0, \quad \int_{(0,\infty)} \frac{d\mu(s)}{1 + s} < \infty.$$

For bounded positive operators the integral converges in norm: its integrand is bounded by $\min\{1, \|A\|/s\}$, which is dominated by a constant times $(1 + s)^{-1}$. The identity function is admissible. Integrating the resolvent estimates against the positive measure and using the triangle inequality therefore proves (897.12); the scalar constant $f(0)X$ cancels. □

For a spectral floor $A, B \ge aI$, apply the theorem to $t \mapsto f(a + t) - f(a)$ and to $A - aI, B - aI$. This gives the sharper increment $k(f(a + \delta) - f(a))$ whenever f is operator monotone on $[a, \infty)$ and continuous at a. In particular it recovers the fractional-power and logarithmic refinements above. No statement for arbitrary unitarily invariant norms is being inferred from this argument.

References

[1] T. A. Loring and F. Vides, *Estimating Norms of Commutators*, Experimental Math. **24** (2015), 106–122. arXiv:1301.4252v3 (2013), Conjecture 1.6 and Lemma 1.7.

[2] D. Herrera, *On the commutator modulus of continuity for operator monotone functions*, preprint, November 2023. arXiv:2311.17448, equation (7), Conjectures 1.8–1.9 and Theorems 1.11–1.12.

[3] MColbrook/AIM, *AIM 897: Pedersen's sharp fractional-power commutator inequality*, repository statement at revision `5d7e26e8444e`. https://github.com/MColbrook/AIM/blob/5d7e26e8444eb98de91988454564e45e011107c4/problems/897-pedersen-s-sharp-fractional-power-commutator-inequality.md.

[4] P. Chansangiam, *Integral Representations and Decompositions of Operator Monotone Functions on the Nonnegative Reals*, preprint, 2013. arXiv:1304.7936, Proposition 1.1.

[5] R. Bhatia, *Matrix Analysis*, Graduate Texts in Mathematics **169**, Springer, 1997, Chapter X, Problem X.5.3, p. 317, and chapter notes, p. 321. doi:10.1007/978-1-4612-0653-8.